

Math 113 – Homework 1 Answers

All assignments must be **turned in on Gradescope** by the deadline. Late assignments are not accepted.

Have questions? Work with your peers, attend office hours, check out the [Student Learning Center](#), and ask in discussion section!

Problems for credit. Turn in these problems; some subset of them will be graded.

1. Fraleigh Exercise 0.6
2. Fraleigh Exercise 0.16
3. Fraleigh Exercise 0.30
4. Fraleigh Exercise 2.2
5. Fraleigh Exercise 2.8
6. Fraleigh Exercise 3.2
7. Fraleigh Exercise 3.30

Practice problems. The following problems will not be graded, and should not be turned in, but are recommended as good practice problems. Solutions to most odd-numbered exercises are available in the back of the textbook.

- Fraleigh Exercises 0.5, 0.9, 0.11, 0.17, 0.25, 0.28, 0.29, 0.31, 0.36
- Fraleigh Exercises 2.3, 2.17, 2.26, 2.28, 2.37
- Fraleigh Exercises 3.3, 3.4, 3.24, 3.25, 3.31, 3.33

1 Expanded problems for credit

Problem 1**Solution notes**

We consider what this set is. $\{n \in \mathbb{Z} \mid n^2 < 0\}$ describes all integers where $n^2 < 0$.

Notice by the trivial inequality $x^2 \geq 0$ for $x \in \mathbb{R}$. Thus, no integers satisfy this condition and it describes the empty set.

Problem 2**Solution notes**

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1. The empty set has only one subset, $\emptyset$. Thus, its power set is $\{\emptyset\}$ with cardinality 1
 2. The power set of $\{a\}$ is $\{\emptyset, \{a\}\}$ so its cardinality is 2.
 3. The power set of $\{a, b\}$ is $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ with cardinality 4
 4. The power set of $\{a, b, c\}$ is $\{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$ with cardinality 8

Problem 3**Solution notes**

For the pair $(0, 3) \in \mathbb{R} \times \mathbb{R}$ for $\geq$ to be an equivalence relation, the symmetric property would have to hold, but notice $0 \not\geq 3, 3 \geq 0$, so it can't be an equivalence relation.

Problem 4**Solution notes**

We have $(a * b) * c = (b * c) = a$.

For the other, we have $a * (b * c) = a * a = a$.

You can not say on the basis of this one calculation, it would have to hold over all possible values associativity holds.

Problem 5**Solution notes**

We have $a * b = ab + 1$ and $b * a = ba + 1$, so $a * b = b * a$, and commutativity holds.

We have

$$a * (b * c) = a * (bc + 1) = a(bc + 1) + 1 = abc + a + 1$$

and we have

$$(a * b) * c = (a * b + 1) * c = (a * b + 1)c + 1 = abc + c + 1$$

which are not equal when $a \neq c$, so it is not associative.

Problem 6**Solution notes**

We first find $\phi(a + b) = -(a + b) = -a - b = \phi(a) + \phi(b)$ which establishes ϕ as a homomorphism

To prove that the function $\phi(a) = -a$ is injective, we see that

$$\phi(a) = \phi(b) \iff -a = -b \iff a = b$$

To prove surjectivity, we see that

$$\forall x \in \mathbb{Z}, \phi(-x) \in \mathbb{Z} \text{ and } \phi(-x) = x$$

and if $x \in \mathbb{Z} \iff -x \in \mathbb{Z}$ Thus, it satisfies the homomorphism condition, injectivity, surjectivity making it an isomorphism

Problem 7**Solution notes**

Suppose we have a binary structure $(S, *)$ and there is an isomorphism $\phi : S \rightarrow S'$ mapping $(S, *)$ to $(S', *')$.

Subject to the conditions of homomorphism, we have that $\phi(a * b) = \phi(a) *' \phi(b)$, and along with this, we're given the condition of associativity and homomorphism for $(S, *)$ which yields

$$(a * b) * c = a * (b * c) \rightarrow \phi((a * b) * c) = \phi(a * (b * c))$$

The LHS is

$$\phi((a * b) * c) = ((\phi(a) * \phi(b)) *' \phi(c))$$

and the RHS is

$$\phi(a * (b * c)) = \phi(a) *' (\phi(b) * \phi(c))$$

We now apply the homomorphism again on two variables, namely $\phi(a * b) = \phi(a) *' \phi(b)$

Substituting yields

$$(\phi(a) *' \phi(b)) *' \phi(c) = \phi(a) *' (\phi(b) *' \phi(c))$$

which are equal, because the original LHS and RHS were identical

To conclude, the last property of isomorphism is surjectivity, and so notice for arbitrary

$$a', b', c' \in S', \exists a, b, c \in S : \phi(a) = a', \phi(b) = b', \phi(c) = c'$$

and so we can substitute the equation above using this mapping.

$$(x' *' y') *' z' = x' *' (y' *' z') \text{ proving associativity} \quad \square$$

2 Practice problems

Practice problems (not for submission):

Fraleigh Exercises 0.5, 0.9, 0.11, 0.17, 0.25, 0.28, 0.29, 0.31, 0.36;
2.3, 2.17, 2.26, 2.28, 2.37; and 3.3, 3.4, 3.24, 3.25, 3.31, 3.33.